Categorical logic

On course webpage:

- notes
- links to videos
- external material

Assesment

- presentations in the seminar slot
- oral exam

Algebraic theories

Perspectives

- Algebraic structure on sets vs. algebraic structure on „spaces"
  In what generality can we interpret algebraic structure
- When are two notions of algebra equivalent
- Categorification: presenting mathematical structures thru the lens of chategory theory
  E.g. algebraic theories as lawvere theories
  categories of algebras as functor cathegory (functional semantics)

A topological group is a ~~set~~ top. space G with continious $m: G \times G \to G$ (topological product),

$e \in G$ $(G^0 \to G)$, $i: G \to G$

$m(x, m(y,z)) = m(m(x,y), z)$

$m(e, x) = x = m(x, e)$

$m(i(x), x) = m(x, i(x)) = e$

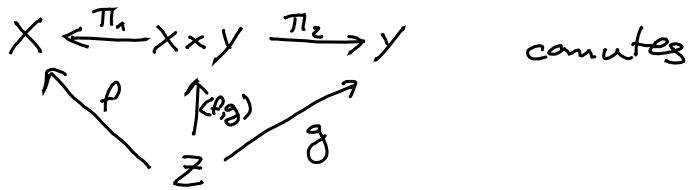
More generaly we can define a group object in a chategory $\mathbb{C}$ in any chategory $\mathbb{C}$ with finite products

A chategery with final products has

- binary products: for any ob X, y we have

$$X \xleftarrow{\pi_1} X \times y \xrightarrow{\pi_2} y$$

such that for any span $(X \xleftarrow{f} z \xrightarrow{g} y)$ there exists $\exists!\ z \xrightarrow{(f,g)} X \times y$ such that



commutes

- terminal object 1
for any object $X \in \mathbb{C}$ there exists a unique map from $X \to 1$

A group object in a category $\mathbb{C}$ with finite products is an object G plus morphisms

$$m : G \times G \longrightarrow G$$

$$e : 1 \longrightarrow G$$

$$i : G \longrightarrow G$$

satisfying

$$\begin{array}{ccccc} G\times(G\times G) & & \xrightarrow{G\times m} & & G\times G \\ \downarrow \cong & & & & \downarrow m \\ (G\times G)\times G & \xrightarrow[m\times G]{} & G\times G & \xrightarrow[m]{} & G \end{array}$$

where $\cong$ is $((\pi_1, \pi_1\circ\pi_2), \pi_2\circ\pi_2)$

$$\begin{array}{ccc} G & \xrightarrow{(e\circ !_G,\, id_G)} & G \\ & \searrow^{id_G} & \downarrow \\ & & G \end{array}$$

+2 for inverse

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{z} & \\ & & \end{array}$$

$$Z\times X \xrightarrow[Z\times f]{(\pi_1, f\circ\pi_1)} Z\times Y$$

The category $\mathrm{Grp}_{\mathbb{C}}$ of group objects in $\mathbb{C}$ is given by:

- Objects: Group objects in $\mathbb{C}$ (G, m, e, i)
- Morphisms $(G, M, e, i) \xrightarrow{h} (G', M', e', i')$ in $\mathbb{C}$ such that

$$\begin{array}{ccc} G \times G & \xrightarrow{h \times h} & G' \times G' \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle m} \\ G & \longrightarrow & G' \end{array}$$

$$\begin{array}{ccc} & 1 & \\ {\scriptstyle e}\swarrow & & \searrow{\scriptstyle e'} \\ G & \xrightarrow[h]{} & G' \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{h} & G' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i'} \\ G' & \longrightarrow & G' \end{array}$$

- identities $(G, m, i, e) \xrightarrow{id_G} (G, m, i, e)$
- composition $(G, m, i, e) \xrightarrow{h} (G', m', i', e') \xrightarrow{h'} (G'', m'', i'', e')$, with $h' \circ h$ in $\mathbb{C}$

An algebraic signature Σ is a collective of operations, where each operation f has an associated arity $k \in \mathbb{N}$ ($\mathbb{N}$ includes 0)

The terms T are built by:

- variables x are terms
- if $t_1 \dots t_k$ are terms and f an operation of arity k, then $f(t_1 \dots t_k)$ is a term.

Notation $t(x_1, \dots, x_n)$ to mean all variables in t contained in the ~~left~~ set $\{x_1 \dots x_n\}$

Equation: A pair of terms (t, t') written

$$t = t_1 \quad (x_1, \dots, x_n)$$

Algebraic theory $T = \Sigma + \mathcal{E}$

(Σ: signature; $\mathcal{E}$: collection of equations)

A T-object in $\mathbb{C}$ (with final product) is:

- an object A of $\mathbb{C}$
- for every operation f in $f \in \Sigma$ of arity k we have a morphism $A^k \xrightarrow{f^A} A$

$$A^0 := 1$$
$$A^1 := A$$
$$A^2 := A \times A$$
$$A^{n+1} = A^n \times A$$

such that all equations in $\mathcal{E}$ are true in A

We interpret any term $t\ (x_1 \dots x_n)$ ($x_1 \dots x_n$ distinct) as a morphism $A^n \xrightarrow{t^A} A$

$$A^n \xrightarrow{x_i} A := A^n \xrightarrow{\pi_i} A$$

$$A^n \xrightarrow{f(t_1 \dots t_k)^A} A := A^n \xrightarrow{(t_1^A \dots t_k^A)} A^k \xrightarrow{f^A} A$$

This gives us an interpretation of all terms

An equation $t = t'$ is satisfied if

$$A^n \xrightarrow{t^A} A = A^n \xrightarrow{(t')^A} A$$

The Category $\mathrm{Mod}(T, \mathbb{C})$ of T-objects in $\mathbb{C}$ is defined by

- Objects: T-objects in $\mathbb{C}$ $(A, (f^A)_{f \in \Sigma})$
- Morphisms $(A, (f^A)_{f\in\Sigma}) \to (B, (f^B)_{f\in\Sigma})$ morphisms $A \xrightarrow{h} B$ s.t. $\forall f \in \Sigma$ of arity k

$$\begin{array}{ccc} A^k & \xrightarrow{h^k} & B^k \\ {\scriptstyle f^A}\downarrow & & \downarrow{\scriptstyle f^B} \\ A & \xrightarrow[h]{} & B \end{array}$$

The Lawvere theory $\mathbb{T}$ of an algebraic theory $T = (\Sigma, \mathcal{E})$

$\mathbb{T}$ objects: $\underline{n}$ $n \in \mathbb{N}$

morphisms: $\underline{m} \longrightarrow \underline{n}$ $\qquad \underline{n} = \underline{1}^n$

are n-tuples of terms where $([t_1], \ldots [t_n])$ ← equivalence classes of terms modulo provable equality from equations

$t_1, \ldots t_n \quad (x_1 \ldots x_m)$

identity $\underline{n} \longrightarrow \underline{n} := \underline{n} \xrightarrow{([x_1] \ldots [x_n])} \underline{n}$

composition $\underline{l} \xrightarrow[(x_1 \ldots x_l)]{([t_1], \ldots [t_m])} \underline{m} \xrightarrow[(x_1 \ldots x_m)]{([t_1'], \ldots [t_n'])} \underline{n}$

$$\underline{l} \xrightarrow[(x_1, \ldots x_l)]{([t_1'[t_1 \ldots t_m / x_1 \ldots x_m]], \ldots)} \underline{n}$$

Two algebraic theories T, T' are equivalent if corresponding lawvere theories are equivalent

- category
- equivalence of category
- functors
- natural transformations

$\mathbb{T}$ has final product:

Terminal object $\underline{0}$

$\underline{m} \times \underline{n} = \underline{m+n}$

Theorem: $Mod(\mathbb{T}, \mathbb{C}) \simeq FP[\mathbb{T}, \mathbb{C}]$

(FP: final product preserving functors)

objects:

arrows: natural transformation

A functor $\mathbb{C} \longrightarrow \mathbb{D}$ (categories with final product) preserves final product if
- $F(1_{\mathbb{C}})$ is terminal object
- $F(X \times Y)$

($F(1_{\mathbb{C}}) = 1_{\mathbb{D}}$

$F(X \times Y) = F(X) \times F(Y)$ ... preserving product on the nose)

$FX \xleftarrow{F(\pi_1)} F(X \times Y) \xrightarrow{F(\pi_2)} FY$ is a product diagram in $\mathbb{D}$

Today

- simply typed λ-calculus interpreted in cartesian closed categories

Cartesian-closed category $\mathbb{C}$

- has terminal product
- any two objects A, B have a product $A \times B$ in $\mathbb{C}$
- any two objects X, Y have exponential Y^X

exponential object has a structure:

- $Y^X \times X \xrightarrow{\epsilon} Y$ "epsilon" "eval"

such that for any map $Z \times X \xrightarrow{f} Y$ there exist a unique map $Z \xrightarrow{\Lambda f} Y^X$ satisfying

$$\begin{array}{ccc} Y^X \times X & \xrightarrow{\epsilon} & Y \\ {\scriptstyle \Lambda f \times X} \uparrow & \nearrow {\scriptstyle f} & \\ Z \times X & & \end{array}$$

Exponentials are unique up to isomorphisms.

Suppose $(X \Rightarrow Y) \times X \xrightarrow{\epsilon} Y$

$(X \Rightarrow' Y) \times X \xrightarrow{\epsilon'} Y$

are both exponentials

then $(X \Rightarrow' Y) \underset{\Lambda' \epsilon}{\overset{\Lambda \epsilon'}{\rightleftarrows}} (X \Rightarrow Y)$

is a pair and is inverse

Examples of cartesian closed categories (ccc)

- Set
- Boolean algebra (More generaly Heyting algebras)
- Cat (The category of categories

An object X is exponentiable, if for any $Y \in |\mathbb{C}|$, the exponential Y^X exists

In Top a space X is exponentiable $\Leftrightarrow$ it iss core compact ($\mathcal{O}(X)$ is a continuous lattice (mreža)

- convinient categories of topological spaces

if X is exponentiable then $(-)^X : \mathbb{C} \to \mathbb{C}$ is a functor

$$\begin{array}{ccc} Y & \longrightarrow & Y^X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \Lambda(f \circ \epsilon_{X,Y})} \\ Y' & \longrightarrow & Y'^X \end{array}$$

X is exponentiable $\Leftrightarrow$ the functor $(-)\times X$ has a right adjoint ($G : \mathbb{C} \to \mathbb{D}$ functor) so that for all object C, D

$$hom_{\mathbb{C}}(FD, C) \cong hom_{\mathbb{D}}(D, GC)$$

Simply-typed λ-calculus

Base : a set of <u>base types</u>

types $A, B, \ldots ::= a \in Base \mid A \times B \mid 1 \mid A \to B$

terms $\Gamma \vdash t : A$

A context Γ is $x_1 : A_1, \ldots, x_k : A_k$ $(k \geq 0)$ such that $i \neq j \Rightarrow x_i \neq x_j$

$$\frac{}{x_1 : A_1, \ldots x_n : A_n \vdash x_i : A_i}$$

Equality relation

$$\frac{\Gamma \vdash t : A}{\Gamma \vdash t = t : A} \qquad \frac{\Gamma \vdash s = t : A}{\Gamma \vdash t = s : A} \qquad + \text{transitivity}$$

congruence relation

$$\frac{\Gamma \vdash s = s' : A \quad \Gamma \vdash t = t' : B}{\Gamma \vdash (s, t) = (s', t') : A \times B} \qquad \frac{\Gamma, x : A \vdash t = t' : B}{\Gamma \vdash \lambda x : A . t = \lambda x : A . t' : A \to B}$$

β-rule

$$\frac{\Gamma \vdash t_1 : A_1 \quad \Gamma \vdash t_2 : A_2}{\Gamma \vdash \pi_i (t_1, t_2) = t_i : A_i}$$

$$\frac{\Gamma, x:A \vdash s : B \qquad \Gamma \vdash t : A}{\Gamma \vdash (\lambda x:A.s)\, t = s[t/x] : B}$$

η-rules

$$\frac{\Gamma \vdash u : 1}{\Gamma \vdash u = * : 1} \qquad \frac{\Gamma \vdash u : A \times B}{\Gamma \vdash u = (\pi_1 u, \pi_2 u) : A \times B}$$

$$\frac{\Gamma \vdash u : A \to B}{\Gamma \vdash u = \lambda x:A.\, u\, x : A \to B} \; x \notin \Gamma$$

λ-theory $T(\text{Base}, \Sigma, \mathcal{E})$

- Base is a set of base types
- Σ a signature with elements $c:A$ such that each c in Σ has exactly one type

  Add rule $\dfrac{}{\Gamma \vdash c:A}\ c:A \in \Sigma$

  Γ constant

- $\mathcal{E}$ is a set of types equations $\Gamma \vdash s=t:A$ between terms $\Gamma \vdash s:A$ $\Gamma \vdash t \vdash A$

  Add rule $\dfrac{}{\Gamma \vdash s=t:A}\ \Gamma \vdash s=t:A \in \mathcal{E}$

Examples

every algebraic theory gives as λ-theory

Base: $\{a\}$

for Every operation of arity k we have

$c_f : a \times \dots \times a \to a$

for every equation ~~we have~~ $t=t'$ with variables $x_1, \dots, x_k$ $x_1:a, x_n:a \vdash t_1 = t = t' : a$

Example (Gödels system T)

Base ($\underline{\mathbb{N}}$)

Signature $0 : \mathbb{N}$, $suc : \mathbb{N} \to \mathbb{N}$, $\{R_A : A \to (A \to A) \to (\mathbb{N} \to A)\}$ (recursor)

Equations $\mathcal{E}$: $x:A \; f:A \to A \vdash R_A \, x \, f \, 0 = x$

$x:A \; f:A, n:\mathbb{N} \vdash R_A \, f \, (suc\, n) = f \, (R_A \, x \, f \, n) : A$

A T-model in a CCC $\mathbb{C}$ is given by

- An in $a^{\mathbb{C}} \in |\mathbb{C}|$ for every $a \in Base$ (objekti)

  This determines interpretation $A^{\mathbb{C}}$ of any type A

$$(A \times B)^{\mathbb{C}} := A^{\mathbb{C}} \times B^{\mathbb{C}}$$

$$1^{\mathbb{C}} := \underline{1}$$

$$(A \to B)^{\mathbb{C}} := (B^{\mathbb{C}})^{A^{\mathbb{C}}}$$

- A typed constant $c : A \in \Sigma$ as arrows

$$\underline{1} \xrightarrow{c^{\mathbb{C}}} A^{\mathbb{C}}$$

Context interpreted by

$$()^{\mathbb{C}} := \underline{1}$$

$$\Gamma, x:A := \Gamma^{\mathbb{C}} \times A^{\mathbb{C}}$$

In general a term $\Gamma \vdash t : A$ will be interpreted as a morphism $\Gamma^{\mathbb{C}} \xrightarrow{[\![\Gamma \vdash t : A]\!]^{\mathbb{C}}} A^{\mathbb{C}}$

This involves a lengthy definition

- subject to the condition that every equation $\Gamma \vdash_T s = t : A \in \mathcal{E}$ (every equation provable in T) holds $(\Gamma \vdash s : A)^{\mathbb{C}} = (\Gamma \vdash t : A)^{\mathbb{C}}$

Theorem: Soundness and completeness

following are equiv.

1) $\Leftrightarrow$ $\Gamma \vdash_T s = t : A$

2) $\Leftrightarrow$ $\forall\, \mathbb{C} \in CCC$ $\forall$. T model in $\mathbb{C}$

$$(\Gamma \vdash s : A)^{\mathbb{C}} = (\Gamma \vdash t : A)^{\mathbb{C}} : \Gamma^{\mathbb{C}} \longrightarrow A^{\mathbb{C}}$$

(every derivable equation in the model is provable and the other way around)

1) $\Rightarrow$ 2) is soundness

2) $\Rightarrow$ 1) is completeness

Today: Categorical algebra (from lecture 1

Monads and their algebras

T-algebra $(A, (m^A)_{m \in \Sigma})$ in any $\mathbb{C}$ with finite product

The category $Alg_T(\mathbb{C})$ of T-algebras in $\mathbb{C}$ and their homomorphisms

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Free Algebra, T-algebra, MONAD

Give an object $X \in |\mathbb{C}|$

The Free T-algebra in $\mathbb{C}$ over X (if it exists)

is given by $(FX, (f^{FX})_{f \in \Sigma})$ a T-algebra

- FX: free algebra over X (object)
- $(f^{FX})_{f \in \Sigma}$: operation (a map)

together with a map $X \xrightarrow{\eta_X} FX$

(insertion generations

such that, for any other pair $(A, (f^A)_{f \in \Sigma})$

((T-algebra + $X \to A$.))

there exists a unique homomorphism

$h: (FX, (f^{FX})_f) \xrightarrow{f} (A, (f^A)_f$

such that

$$\begin{array}{ccc} FX & \xrightarrow{h} & A \\ {\scriptstyle \eta}\uparrow & \nearrow_{g} & \\ X & & \end{array}$$

commutes

Free group over set X of generators exists and is given as the set of irreducible words over a set $X \sqcup X^{-1}$ with insertion of generators $x \in X \mapsto x$ (the 1-symbol word)

Free $(\Sigma, \mathcal{E})$-algebras in <u>Set</u> are constructed by

$FX :=$ the set of terms with no variables over $\Sigma \sqcup \{c_x ; x \in X\}$ / derivable equality from equations in $\mathcal{E}$??

(new constant for each x)

$$f^{FX}([t_1], \ldots, [t_k]) := [f(t_1 \ldots t_k)]$$

$$\eta_X(x) := [c_x]$$

<u>Set</u> has free $(\Sigma, \mathcal{E})$-algebras

$\mathbb{C}$ has free $(\Sigma, \mathcal{E})$-algebras if it has free algebras over every object X

Suppose $\mathbb{C}$ has free T-algebras, suppose we have a choice of $(FX, (f^{FX}))$ and η_X for every X.

Then F defines a functor $\mathbb{C} \longrightarrow Alg_T(\mathbb{C})$ (the free algebra functor)

$$\begin{array}{ccc} X & & (FX, (f^{FX})_f) \\ f\downarrow & \longmapsto & \downarrow h \text{ unique s.t.} \\ Y & & (FY, (f^{FY})_f) \end{array} \quad \text{commutes}$$

We see that η is a natural transformation $Id_{\mathbb{C}} \to UF$

$$\begin{array}{ccc} FX & \overset{h}{\dashrightarrow} & FY \\ {\scriptstyle \eta_X}\uparrow & & \uparrow{\scriptstyle \eta_Y} \\ X & \xrightarrow{f} & Y \end{array}$$

We also have an obvious functor

$U : Alg(\mathbb{C}) \longrightarrow \mathbb{C}$ forgetful functor

F is left adjoint to U

Proposition: $F \dashv U : Alg(\mathbb{C}) \longrightarrow \mathbb{C}$

F is left adjoint to U if

$F : \mathbb{C} \longrightarrow \mathbb{D}$ is left adjoint to $G : \mathbb{D} \longrightarrow \mathbb{C}$ (notation $F \dashv G : \mathbb{D} \longrightarrow \mathbb{C}$).

if there exist bijections

$\Psi_{xy} : \mathbb{D}(Fx, y) \xrightarrow{\cong} \mathbb{C}(x, Gy)$

natural in x and y

Proof

$$Alg_T(\mathbb{C})\big((FX, (f^{FX})f), (A, (f^A)f)\big) \xrightarrow{\cong} \mathbb{C}(X, A)$$

$$(FX, -) \xrightarrow{h} (A, -) \longmapsto h \circ \eta_X$$

unique s.t. $\begin{array}{ccc} FX & \xrightarrow{h} & A \\ {\scriptstyle \eta_X}\uparrow & \nearrow_{g} & \\ X & & \end{array}$ $\longleftarrow g$

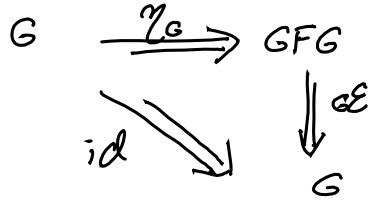
Alternative definition for adjunction

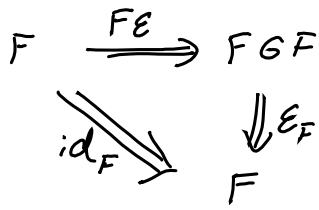
There exist natural transformations

$\eta : id_{\mathcal{C}} \Rightarrow FG$ (unit)

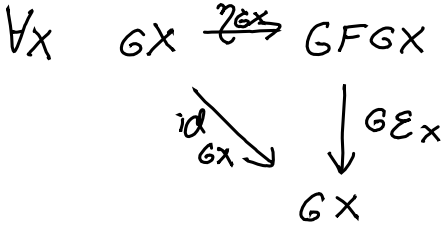
$\varepsilon : FG \Rightarrow id_{\mathcal{D}}$ (counit)

such that





natural transformation



The unit is $X \xrightarrow{\eta_X} UFX$

Exercise: Workout the counit

We have (Σ, E) such that $F \dashv U : Alg_T(\mathbb{C}) \to \mathbb{C}$

$T := UF \qquad T : \mathbb{C} \to \mathbb{C}$

$\eta : Id_{\mathbb{C}} \Rightarrow T$ unit

$\mu : T^2 \Rightarrow$ multiplicative

μ defined as follows for any object $X \in |\mathbb{C}|$

$T^2X \xrightarrow{\mu_X} TX \qquad UFUFX \to UFX$

$FFX \to FX$

$FX \xrightarrow{id_{FX}} FX$ there is a unique

homomorphism $(FFX, (A^{FFX})) \xrightarrow{\mu_X} (FX, (A)^{FX})$

$$\begin{array}{ccc} F^2X & \xrightarrow{\mu_X} & FX \\ & {\scriptstyle \eta_{FX}} \nwarrow \quad \nearrow {\scriptstyle id_X} & \\ & FX & \end{array}$$

This indeed defines a natural transformation

In the case of $(\Sigma, \mathcal{E})$ on Set

$\mu : FFX \longrightarrow FX$ (FX equivalence classes $[t]$ of terms t over $\Sigma \sqcup \{c_x\}_X$)

FFX equivalence classes $[T]$ of terms t over $\Sigma \sqcup \{c_{[t]}\}_{[t] \in FX}$

$\mu_X [c_{[t]}] := [t]$

$\mu_X [f([t_1], \ldots, [t_n])] := [f(t_1', \ldots, t_n')]$ where each t_i' is such that $\mu_X [t_i] = [t_i']$

A **Monad** on any $\mathbb{C}$ is (T, η, μ)

where $T : \mathbb{C} \longrightarrow \mathbb{C}$

unit $\eta : Id_{\mathbb{C}} \Rightarrow T$

multiplication $\mu : T^2 \Rightarrow T$

such that

$$\begin{array}{ccccc} & & T^2X & & \\ & \eta_{TX} \nearrow & \downarrow \mu_X & \nwarrow T\eta_X & \\ TX & \xrightarrow[id_X]{} & T & \xleftarrow[id_X]{} & TX \end{array}$$

$$\begin{array}{ccc} T^3X & \xrightarrow{\mu_{TX}} & T^2X \\ {\scriptstyle T\mu_X}\downarrow & & \downarrow{\scriptstyle \mu_X} \\ T^2X & \xrightarrow[\mu_X]{} & TX \end{array}$$

A monad is just a monoid in the monoidal (with composition as the monoidal product) category $[\mathbb{C},\mathbb{C}]$

An Eilenberg-Moore (EM algebra) for a monad

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